LESSON 6.8

Generate Equivalent Expressions

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use the commutative, associative, and identity properties to rewrite expressions.

WHY THIS MATTERS You're a band director arranging musicians on stage. You have 3 guitarists, 5 drummers, and 2 keyboard players. Whether you seat the guitarists first or the drummers first, you still have 3 + 5 + 2 = 10 musicians. The order you arrange them doesn't change the total!

REMEMBER

Generate Equivalent Expressions

Properties of Operations. Formula: Commutative: a + b = b + a | Associative: (a + b) + c = a + (b + c) Properties are math rules that are always true. The Commutative Property lets you change the order. The Associative Property lets you change the grouping (the parentheses).

1. Commutative Property: Order does not change sum or product.
2. Associative Property: Grouping does not change sum or product.
3. Identity Property: a + 0 = a; a · 1 = a

Words: **Properties of Operations** ***(Propiedades de las operaciones)*** **Commutative Property** ***(Propiedad conmutativa)*** **Associative Property** ***(Propiedad asociativa)*** **Identity Property** ***(Propiedad de identidad)*** **Property** ***(Propiedad)***

1 **Circle the letter of the best answer. Show how you know.**

What is the GCF of 18 and 24?

(A) 2
(B) 3
(C) 6
(D) 72

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

What is the LCM of 4 and 6?

(A) 2
(B) 10
(C) 12
(D) 24

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

One bell rings every 8 minutes and another every 12 minutes. They just rang together. After how many minutes will they ring together again?

(A) 4 minutes
(B) 20 minutes
(C) 24 minutes
(D) 96 minutes

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

You have 30 pencils and 45 erasers to split into identical kits with nothing left over. What is the greatest number of kits?

(A) 3 kits
(B) 5 kits
(C) 15 kits
(D) 90 kits

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A sound engineer needs to calculate 4 × 17 × 25 mentally to find the total cost of audio cables. Using the associative and commutative properties, she rearranges: 4 × 25 × 17 = 100 × 17. Which property lets you change the ORDER of the factors?

(A) Commutative
(B) Associative
(C) Distributive
(D) Identity

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A sound engineer needs to calculate 4 × 17 × 25 mentally to find the total cost of audio cables. Using the associative and commutative properties, she rearranges: 4 × 25 × 17 = 100 × 17. Which property lets you change the GROUPING of the factors?

(A) Commutative
(B) Distributive
(C) Identity
(D) Associative

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How can you tell the difference between the commutative and associative properties? What clue helps you decide?

SENTENCE STARTER The ___ property changes the ___, while the ___ property changes the ___. I can tell by looking at whether the ___ moved or the ___ changed.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Drag each track on the playlist into the property of operations it shows. Ask yourself: did the order change, the grouping change, or did a 0 or 1 keep the value the same?

Groups: Commutative Property | Associative Property | Distributive Property | Identity Property

_____ 2 + 5 = 5 + 2

_____ 5 × 9 = 9 × 5

_____ (2 + 6) + 4 = 2 + (6 + 4)

_____ (3 × 2) × 5 = 3 × (2 × 5)

_____ 5(7 + 2) = 5 × 7 + 5 × 2

_____ 2(x + 5) = 2x + 10

_____ 11 + 0 = 11

_____ 6 × 1 = 6

2 **Write the letter of the matching item on each line.**

Match each playlist track to the property it plays.

_____ 8 × 6 = 6 × 8

_____ (4 × 2) × 3 = 4 × (2 × 3)

_____ 6(2 + 3) = 6 × 2 + 6 × 3

_____ 10 × 1 = 10

_____ 0 + 19 = 19

_____ 9 + 5 = 5 + 9

(A) Commutative (Multiplication)

(B) Associative (Multiplication)

(C) Distributive

(D) Identity (Multiplication)

(E) Identity (Addition)

(F) Commutative (Addition)

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A sound engineer needs to calculate 4 × 17 × 25 mentally to find the total cost of audio cables. Using the associative and commutative properties, she rearranges: 4 × 25 × 17 = 100 × 17. What is 4 × 17 × 25?

(A) 170
(B) 425
(C) 1,000
(D) 1,700

SHOW EACH PARTIAL PRODUCT, THEN ADD THEM

4 **Circle the letter of the best answer. Show how you know.**

Track 2 on the playlist plays 5(7 + 2) = 5 × 7 + 5 × 2. Which property is this?

(A) Commutative Property
(B) Associative Property
(C) Identity Property
(D) Distributive Property

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Track 3 plays (7 + 3) + 5 = 7 + (3 + 5). Only the parentheses moved. Which property is this?

(A) Commutative Property
(B) Distributive Property
(C) Associative Property
(D) Identity Property

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Track 4 plays 1 × y = y. Which property is this?

(A) Distributive Property
(B) Commutative Property
(C) Associative Property
(D) Identity Property

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How can you tell the difference between the commutative and associative properties? What clue helps you decide?

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